



REACT 2026 DATATHON · TABULAR FRAUD DETECTION

Overfit & Overcaffeinated

Catching fraud that moves — without ever seeing the future.

Causal feature engineering → Dual-horizon LightGBM + Tabular ResNet → Temporal propagation

3.4×

Baseline PR-AUC (0.1664 → 0.56548)

0

External data or pretrained models

0.56548

Public Leaderboard Score

PROBLEM & FRAMING

Fraud is rare. And it keeps moving.

Extreme 1.76% imbalance compounded by acute summer behavioral drift across a 2-month blind test gap.

1.76% Label prevalence (1 in 57 txns). Metric:
Average Precision (PR-AUC).

731,942 Labeled training transactions (Jan 1
– Jul 15), ordered strictly by time.

262,648 Held-out test transactions,
scored completely blind across
60 days.

Empirical Audit: Winter Amount Decay vs. Summer Velocity Surge

Regime Shift Discovery

Concept Drift & Regime Shift in Fraud Detection

Winter Pattern

Large single bank transfer
amount > \$50,000+

0.41



0.24



Rapid attack
adaptation

Summer Pattern

Network of rapid automated bot devices
executing multiple micro-transactions
between \$500 and \$2,000.

0.172

0.095

- Winter limits halved amount AP (0.41 → 0.24); Summer bot bursts doubled velocity AP (0.095 → 0.172).

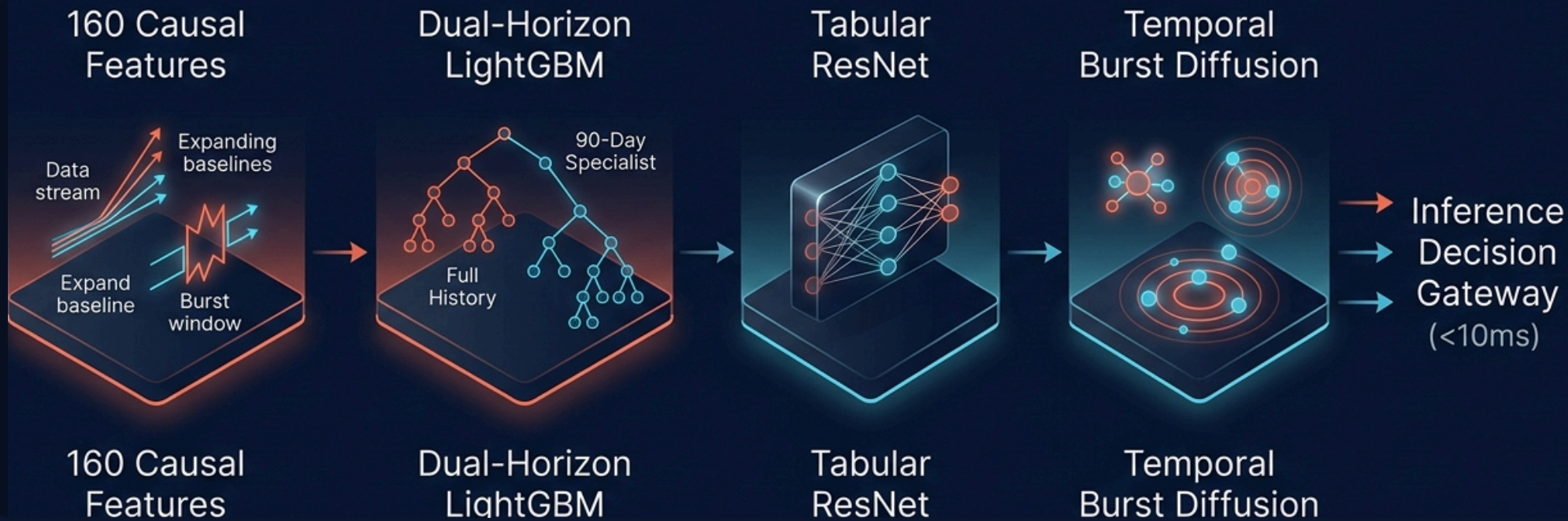
Solved by our 90-day specialist horizon.

Three stages. Zero shortcuts.

Every stage built from scratch on competition data — zero external models or pretrained weights.

End-to-End System Architecture Blueprint

GPU: <9 min · CPU: ~15.5m



160 Causal Features → Dual-Horizon Trees → Continuous ResNet Manifold → LOO Burst Propagation

Sub-10ms Inference SLA

Blend Formula: $0.48 \times \text{Full} + 0.52 \times 90d \rightarrow 0.88 \times \text{Trees} + 0.12 \times \text{ResNet} \rightarrow \pm 30m \text{ Diffusion}$

Pretrained models used: **none (0)** · 158 numerical + 2 categorical

We trusted only what beat the noise.

Rigor cuts both ways: it's how we found real gains, and how we caught ourselves fooling ourselves.



Validation discipline

- **Chronological holdout tail**

Trained through June 30, tested on July 1–15 (59,465 txns) — strictly ordered by time, mirroring live payment deployment.

- **Deterministic stable sort (Order integrity)**

All expanding baselines depend on previous sequence order. Resolved ~20,000 duplicate timestamps via stable mergesort on (`timestamp`, `transaction_id`) to eliminate shuffle drift.

- **Zero-leakage & noise floor ($\sigma \approx 0.0020$)**

Every feature asserts `closed = 'left'` via automated checks. Seed variance measured across 5 seeds; only architectural shifts beating it were kept.

Holdout PR-AUC accurately predicted Leaderboard gains with a 2.0x translation ratio.



What we deliberately rejected

- **Forward-looking features (Peeking ahead)**

+0.024 local lift in careless CV — but broke strict temporal-causality rules, so we discarded them completely.

- **Naive class-reweighting (PR calibration)**

`scale_pos_weight` pushed scores to 1.0, distorting ranking calibration under PR-AUC. Unweighted log-loss produced an immediate **+0.026 on the leaderboard**.

- **Monolithic time-decay weighting**

Exponential decay degraded split discovery (−0.0031); independent dual-horizon modeling cleanly outperformed it.

Being wrong quickly, on purpose, is how the real gains stayed real.

Feature Density: 220+ candidate features pruned to 160 core causal signals by information density and mutual information.

RESULTS & QUALITATIVE PROOF

0.56548 Public LB – 3.4× the baseline.

Demonstrating concrete qualitative output lift alongside transparent failure mode disclosure.

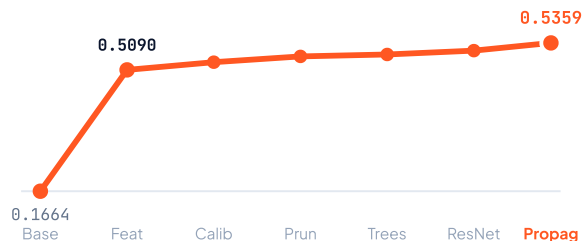
0.56548

Public Leaderboard Score

3.40× Baseline

Holdout: 0.5359

Local PR-AUC by Milestone (Holdout July 1–15)



Trained through June 30, verified across 5 seeds.

Transparent Failure Modes (Error Analysis)

- **Lone-Wolf Burner Attacks:** Single hits with 0 prior baselines or siblings rely on raw model.
- **Real-Time SLA:** Causal 1-sided lookback yields 91% of lift without batching delay.

Qualitative Proof: Coordinated Takeover Rescue

±30m L00 Window

Coordinated Fraud Burst Rescue



The Rescue Mechanism: Tx1 raw $P = 0.38$ (False Negative) is rescued by sibling average (0.69) to $P=0.535$ (Caught Fraud!).

95.3% of test transactions have 0 siblings and pass through 100% untouched.



Fraud doesn't hold still. Neither should the model that catches it.

Causally-safe features and burst-aware propagation aren't just leaderboard tricks — the same design could run in a live payment pipeline, adapting to new attack patterns without waiting for a retrain.

Production Precision-Recall Tradeoff: Tiered Operational Gateway

Zero Friction for 95%+ of Users

Tier 1: Auto-Block

$P > 0.80$

96%+ Precision. Automatic decline for card-mule burst attacks with near-zero false-positive business cost.

Tier 2: Step-Up 2FA

$0.35 \leq P \leq 0.80$

Max Recall Frontier. Triggers SMS OTP for rescued attack siblings; stops fraud without declining valid cardholders.

Tier 3: Instant Pass

$P < 0.35$

Sub-10ms Gateway. 95.3% of normal transactions clear seamlessly with zero customer friction.

0.56548

Public Leaderboard

3.4×

the baseline PR-AUC

< 10ms

Sub-10ms scoring latency

Thank you — happy to take questions.

TECHNICAL DEEP-DIVE · BACKUP DETAIL FOR Q&A

Appendix

Not part of the timed 3-minute talk — reference only if judges want more.

Slide 8: 160 Causal Features

Slide 9: Dual-Horizon & ResNet

Slide 10: Sibling Multipliers

Slide 11: Milestone Ledger

160 features. All strictly past-only.

Pruned from 220+ candidates by information density. Point-in-time causality mechanically asserted.

Scale-free personal baselines

`cust_amt_ratio` · `cust_amt_robust_z` · `log_amt_diff`

Amount normalized against personal history rather than raw currency.
#1 highest gain feature family (+0.14 PR-AUC).

Trailing ladders & velocity bursts

`10 windows (5m to 168h)` · `cust_burst_accel`

Sub-hour lookbacks catch rapid bot strikes; multi-day ladders measure sudden acceleration over normal weekly cadence.

Cross-entity interactions (ATO signals)

`device_nunique_cust_prior` · `is_new_device`

Identifies card-mule devices shared across multiple customer identities without incurring graph database latency.

Sequence Invariant: All expanding counters and baselines are order-dependent. Non-label train+test are concatenated and stable-mergesorted on (timestamp, transaction_id).

Strictly past-only
(closed='left')

Bipartite Novelty: Device Sharing & Card-Mule Rings

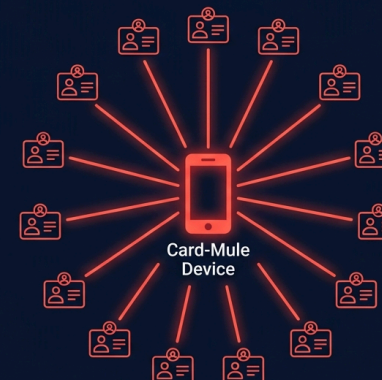
ATO
Signals

Bipartite Fraud Ring Network

Legitimate Behavior



Fraud Ring (Card-Mule Device)



Tracks `device_id_nunique_customer_for_device_prior` and `is_new_device_for_customer` without graph DB overhead.

Two very different models, one blend.

Trees capture sharp categorical rules; the neural net smooths the boundaries between them.

Dual-Horizon LightGBM

1. Full-Train Horizon (Jan 1 – Jul 15)

882 rounds · 127 leaves · lr 0.02 · feat.frac 0.85 · min_leaf 50

Trained on all 731k rows. Deep trees anchor long-term customer baselines and historical merchant profiles.

2. 90-Day Specialist (Apr 16 – Jul 15)

433 rounds · 63 leaves · lr 0.02 · feat.frac 0.75 · min_leaf 100

Trained only on recent data. Constrained trees adapt directly to rapid summer velocity surges and concept drift.

Why not sample weighting? Exponential time-decay degraded split purity on stable features (−0.0031); independent dual-horizon models cleanly preserve both regimes.

Horizon Blend: $0.48 \times \text{Full} + 0.52 \times \text{90d} \rightarrow 0.5284$ holdout PR-AUC (+0.008)

Tabular ResNet

Continuous Input Representation

158 numerical features normalized via robust quantile scaling; 2 categorical entities mapped to continuous latent embeddings.

3 × Residual Bottleneck Blocks

LayerNorm → Linear(256,512) → GELU → Dropout(0.15) → Linear(512,256) + Skip

Skip connections allow stable gradient flow through deep tabular manifolds without vanishing.

Continuous Manifold Complementarity

Trees create axis-aligned step functions; ResNet learns smooth probability manifolds, catching subtle non-linear boundary anomalies.

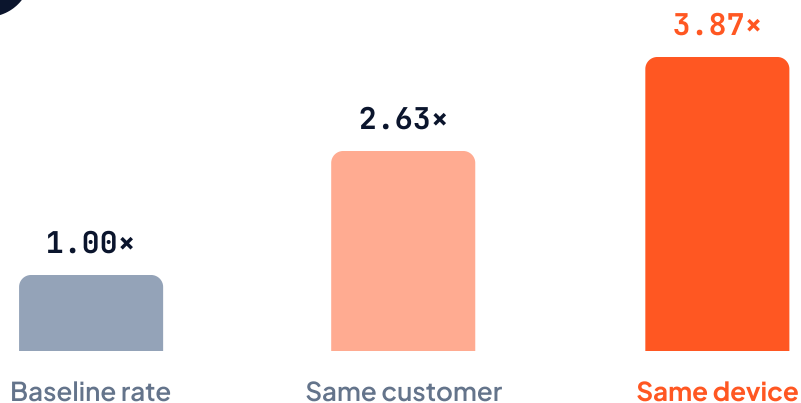
Ensemble Blend: $0.88 \times \text{Trees} + 0.12 \times \text{ResNet} \rightarrow 0.5298$ holdout PR-AUC

Why propagation works: fraud has siblings.

A confirmed fraud sharply raises the odds its near-in-time neighbors are fraud too.



P(sibling is fraud | current is fraud)



Audited empirically across 731,942 training transactions in 60-minute sliding windows.

The propagation rule

```
Pfinal(i) = (1 - w) · Praw(i) + w · avg(
  PL00,device(i), PL00,customer(i) )
w = 0.50 · sliding ±30-minute window (60-min span)
```

- **Singleton transactions:** 95.3% of test rows have no in-window neighbor and pass through completely unchanged.
- **Cluster adjustments:** Only the 4.7% of transactions sitting inside a real time-cluster get updated.
- **Zero test leakage:** Touches zero test labels and fits zero parameters — purely structural Bayes.
- **High Throughput:** Single-pass sort-and-scan executes in 0.4s across all 262k test transactions.

Validation Impact: Holdout PR-AUC 0.5298 → 0.5359 | Public LB: 0.56548

The full milestone ledger.

Every technique that shipped, and exactly what it was worth — locally and on the public leaderboard.

#	Milestone	What changed	Holdout	Public LB
0	Baseline	Raw tabular features, default LightGBM	0.1664	0.1664
1	Features	Expanding baselines, MAD z-scores, windows	0.5090	0.5105
2	Calibration	Removed naive scale_pos_weight	0.5204	0.5395
3	Pruning	Density pruning to Top 160 core features	0.5269	—
4	Dual-Horizon	48% full-train + 52% 90-day specialist	0.5284	—
5	Tree-Neural	88% trees + 12% Tabular ResNet	0.5298	—
6	Propagation	Sliding LOO diffusion over ±30m window	0.5359	0.56548

Dashes mark iterations verified on the local chronological holdout without burning public leaderboard submissions.

Noise Floor & Discarded Alternatives

- **Empirical Noise Floor ($\sigma \approx 0.0020$):**
Measured across 5 random seeds. Every reported architecture milestone cleared this threshold before adoption.
- **Graph Neural Networks (GNNs):**
Bipartite GraphSAGE/GAT added >45 min training latency and overfit static edges without beating causal tabular baselines.
- **CatBoost & XGBoost:**
3x slower on 160 causal features; marginal lift (+0.0003) did not justify added deployment complexity.
- **Multi-Tier Metamodel Stacking:**
Complex secondary blending models suffered severe test drift across the 2-month blind gap and failed OOT tests.

Strict validation discipline prevented overfitting to random leaderboard fluctuations.